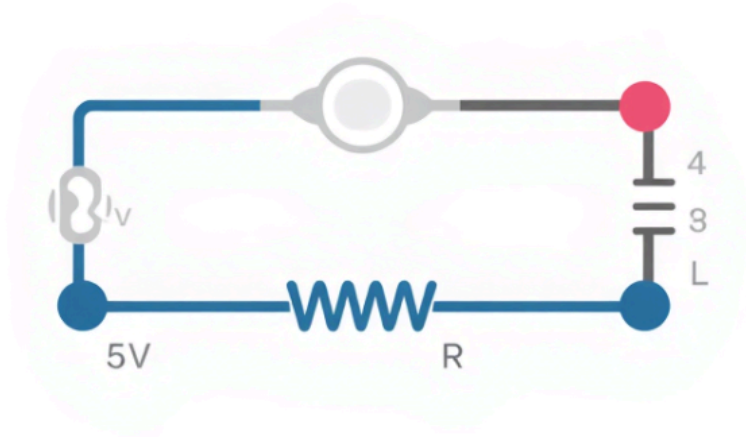
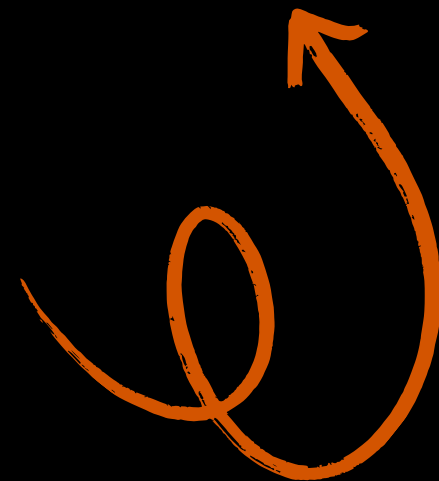


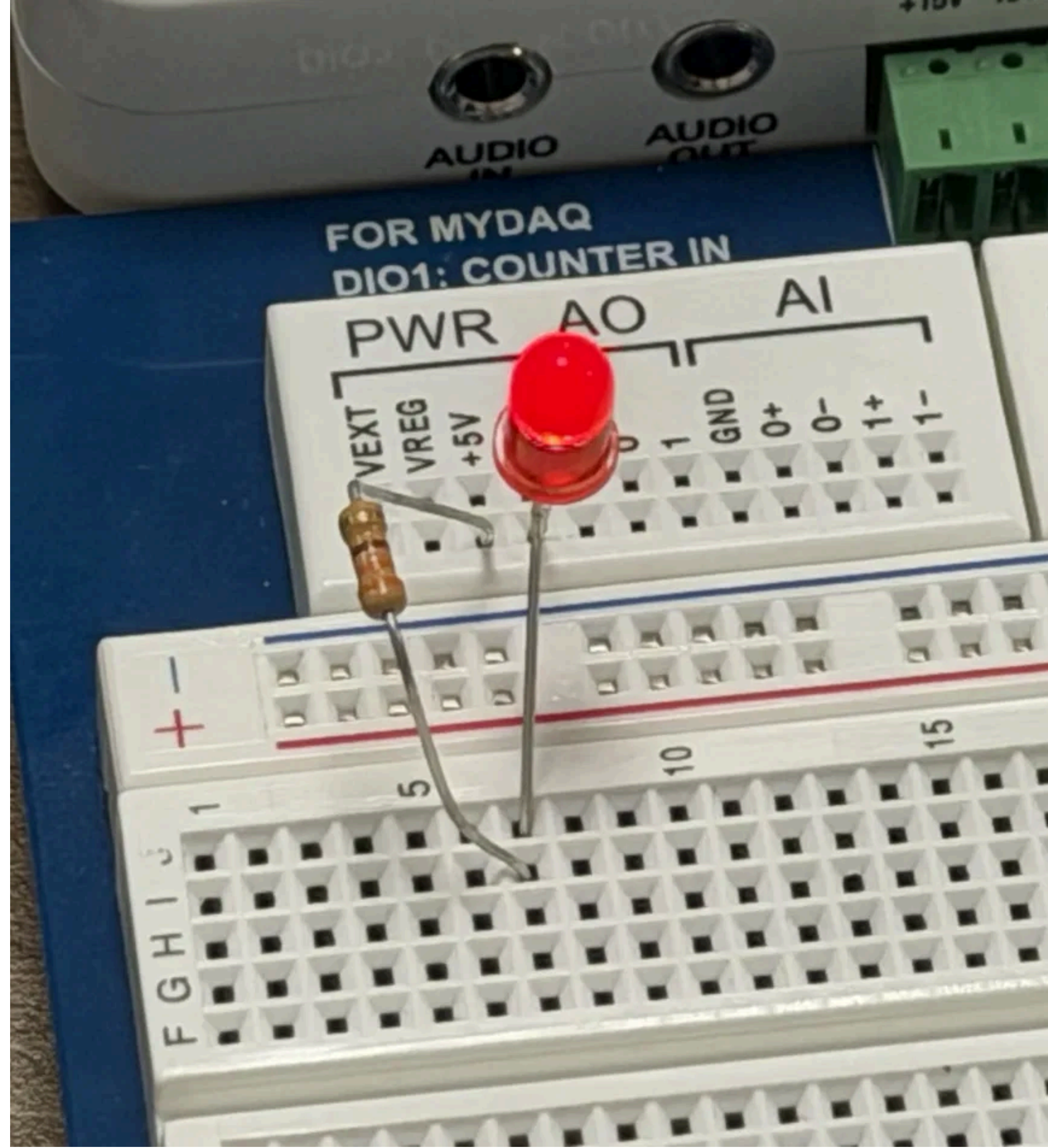
Investigating Basic Circuits

Presented by Conner Glover



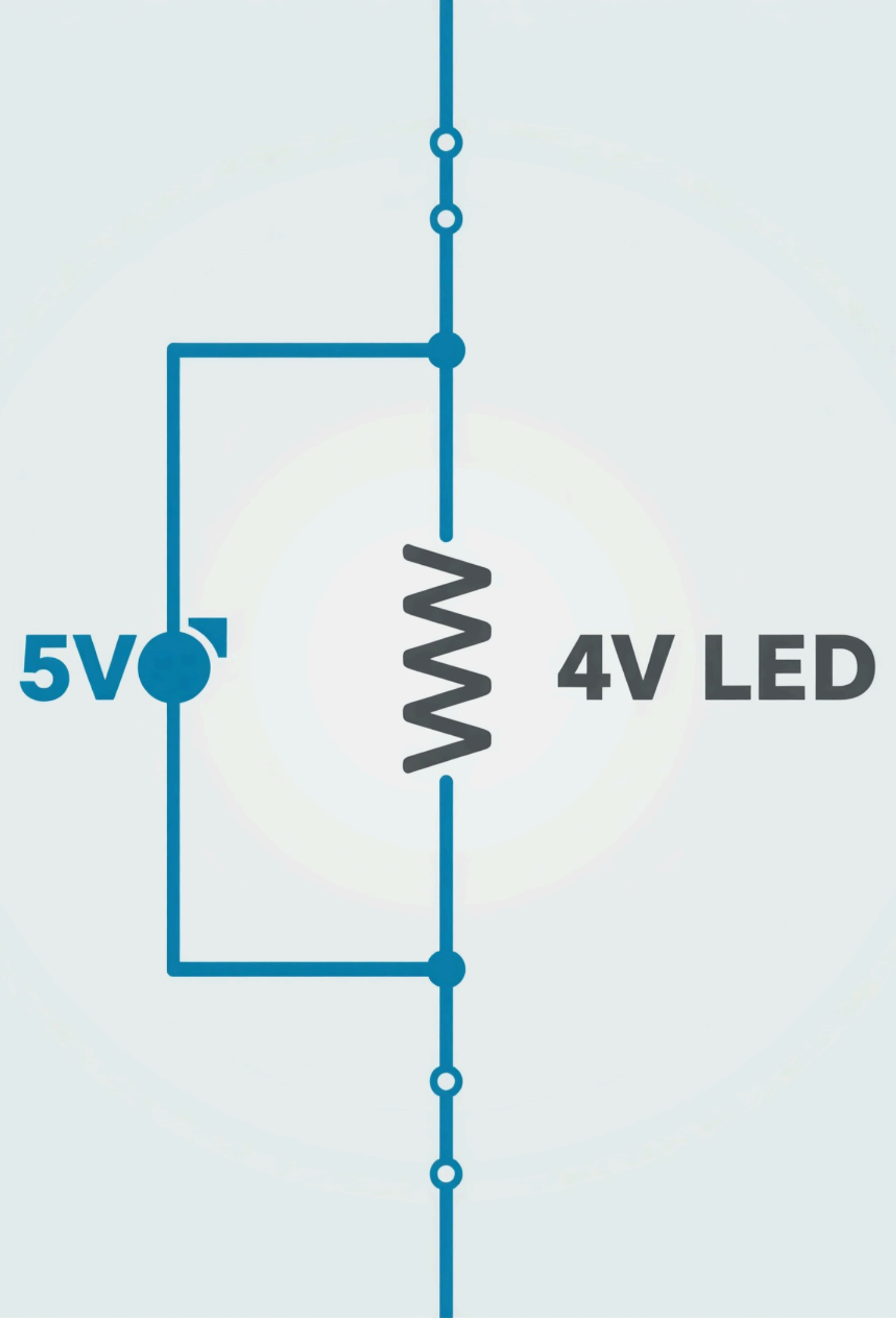
Creating a Simple Series Circuit

This circuit features a **5V power source**, a 470 ohm resistor, and an LED connected in series to ground, demonstrating basic circuit principles and current flow essential for understanding electronics.



Flipping the LED's Orientation

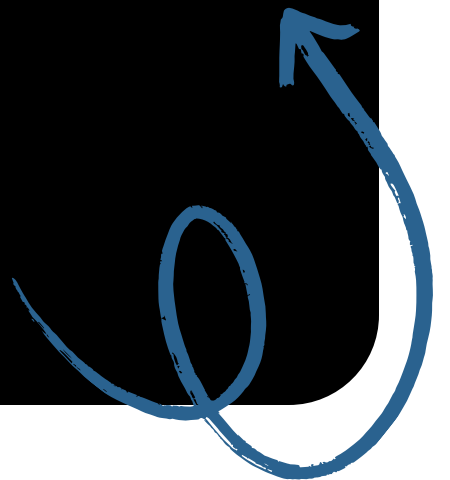
Reversing the LED's orientation demonstrates its **diode behavior**, allowing current to flow in only one direction. This fundamental property is crucial for understanding circuit functionality and electronic design principles.



First DMM Reading

Measuring Voltage in the Circuit

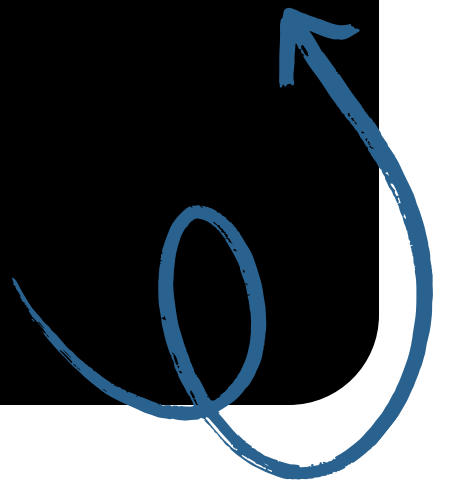
The digital multimeter displayed a reading of **4.8 V DC**, confirming the voltage across the resistor and LED together, illustrating the importance of proper probe orientation for accurate measurements.



Increasing Measurement Precision

Optimizing DMM Settings for Accuracy

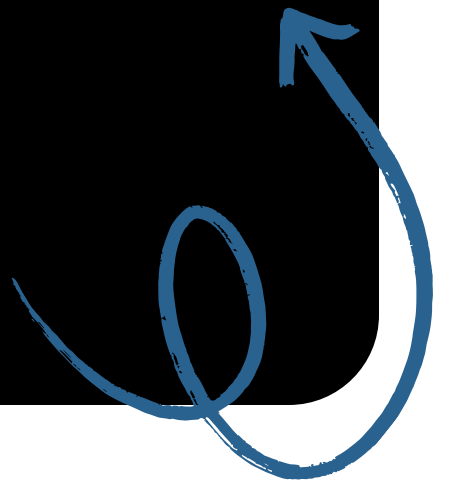
By selecting a smaller voltage range on the digital multimeter, we refined our measurement to 4.76 V DC, enhancing precision and allowing for more accurate circuit analysis.



Range Overload

Understanding DMM Limitations in Circuits

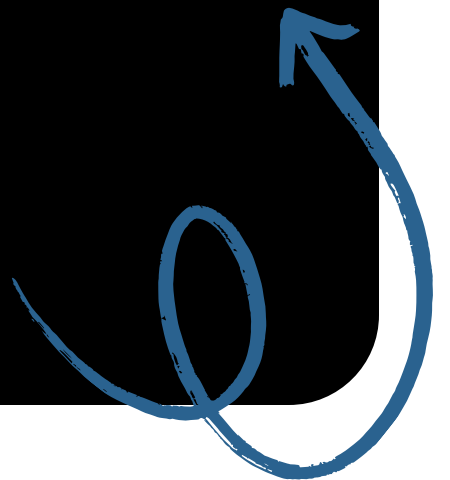
When switching to a smaller voltage range, the DMM displayed "+OVER," indicating the circuit's voltage exceeded this range. This highlights the importance of selecting appropriate ranges for accurate readings.



Voltage Across a Wire

Measuring Resistance in a Circuit

When the DMM probes were placed across the plain wire, it measured **0.00 V DC**, confirming that wires have effectively no resistance and exhibit no voltage drop.



Resistor vs. LED Voltage

Understanding Voltage Drops in Circuits

Voltage Across Resistor

The resistor exhibits a voltage drop of **2.92 V**, confirming its role in limiting current flow within the circuit, illustrating basic principles of resistance and energy conservation.

Voltage Across LED

The LED demonstrates a voltage drop of **1.85 V**, highlighting its unique properties as a diode, allowing current to flow in only one direction and emitting light when forward-biased.

Total Voltage Calculation

The sum of the voltage drops across the resistor and LED totals approximately **4.76 V**, aligning with Kirchhoff's Voltage Law, which states that total voltages in a series circuit must equal the supply voltage.



Applying Ohm's Law

Understanding Current in a Circuit

Resistor Specifications

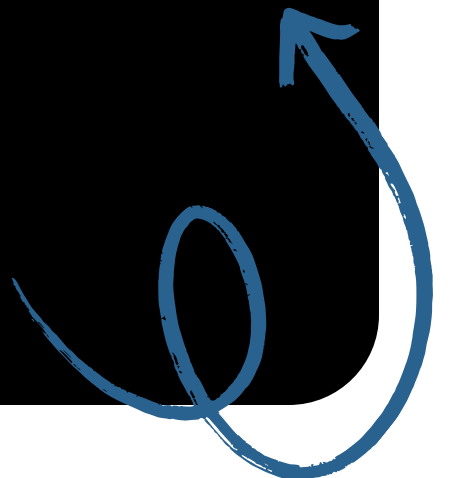
The resistor used in this circuit is rated at **330 ohms**, providing resistance that slows down the current flow, ensuring the LED operates within its safe voltage range.

Voltage Measurement

The measured voltage across the resistor is **2.92 V**, illustrating how voltage is distributed in a circuit and how it can be calculated using Ohm's Law principles.

Calculating Current

By applying Ohm's Law ($V = IR$), the current calculated through the circuit is **8.85 mA**, demonstrating the relationship between voltage, resistance, and current in electrical systems.



Thank you

